

CI/CD Workflow for Selenium Automation Tests

DevelopersHubCorporation Internship | Domain: Software Quality Assurance (SQA)

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1. What is CI/CD?

CI/CD stands for **Continuous Integration** and **Continuous Delivery**. It is a practice where code changes are automatically built, tested, and deployed.

Continuous Integration (CI)

- Developers push code to a shared repository
- Automated builds and tests run on every push
- Issues are caught early

Continuous Delivery (CD)

- After passing tests, code is automatically deployed
- Reduces manual effort and human error

2. Role of QA in CI/CD

Quality Assurance engineers ensure that automated tests are integrated into the CI/CD pipeline. This includes:

- Writing automated test scripts (like Selenium tests)
- Running tests on every code change
- Reviewing test results and reporting failures
- Maintaining test suites for regression testing

3. How My Selenium Tests Would Run in a Pipeline

Below is a sample **GitHub Actions** workflow that would run my Week 3 Selenium tests automatically.

Workflow File: ``.github/workflows/selenium-tests.yml``

```
``yaml
```

```
name: Selenium Automation Tests
```

```
on:
```

```
  push:
```

```
    branches: [ main ]
```

```
  pull_request:
```

```
    branches: [ main ]
```

```
  schedule:
```

```
    - cron: '0 2 * * *' # Runs daily at 2 AM
```

```
jobs:
```

```
  test:
```

```
    runs-on: ubuntu-latest
```

```
    steps:
```

```
      - name: Checkout code
```

```
        uses: actions/checkout@v3
```

```
      - name: Set up Python
```

```
        uses: actions/setup-python@v4
```

```
        with:
```

python-version: '3.11'

- name: Install dependencies

run: |

pip install selenium

pip install webdriver-manager

- name: Run Login Test

run: python Week-3/Automation-Scripts/login_test.py

- name: Run Dashboard Test

run: python Week-3/Automation-Scripts/dashboard_test.py

- name: Upload Test Screenshots

uses: actions/upload-artifact@v3

with:

name: test-screenshots

path: |

login_success.png

sorting_test.png